

Diego Alejandro Rueda Cadavid

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Profile

Environmental Engineer with an M.Sc. in Physics from the National University of Colombia, currently pursuing a Ph.D. in Chemical Engineering at the Federal University of Santa Catarina (UFSC), Brazil. My research focuses on the synthesis and catalytic design of functional materials for hydrogen production and oxygen transport. I have strong experience in the synthesis and characterization of semiconductor perovskites and related nanostructured materials. I am especially interested in applying this background to perovskite-based solar cell research, bridging catalysis, energy conversion, and advanced materials engineering.

Education

Student PhD in Chemical Engineering [Federal University of Santa Catarina](#) Florianópolis, Brazil Sep.2024 - Aug.2028

Focus: New catalytic materials towards to hydrogen production and oxygen transport.

Master's in Sciences-Physics [National University of Colombia](#)

Bogotá D.C. Aug.2022 - Dec.2024

Focus: Physics of materials, solid-state, and modeling of materials for environmental applications.

Master's in Environmental Sciences [University of Cádiz](#)

Cádiz, Spain Sep.2019 - Jul.2021

Focus: Modeling and estimation of greenhouse gases and their radiative effects on the atmosphere.

Environmental Engineering [National University of Colombia](#)

Palmira, Valle del Cauca Aug.2013 - May.2019

Focus: Synthesis and characterization of TiO₂ for catalytic applications of GHG and photocatalysis.

Publications

- Ni-induced lattice distortion and oxygen vacancy engineering in La_{0.5}Ca_{0.5}Mn_{1-x}Fe_{0.5}Ni_xO₃ for visible-light ethanol photoreforming hydrogen production (submitted to *International Journal of Hydrogen Energy*). Diego Rueda, Dachamir Hotza, Sergio Yesid Gómez González.
- Research Trends and Technological Insights on Double Perovskites for Hydrogen Production and Storage: A Bibliometric and Text-Mining Analysis (in preparation). Diego Rueda, Dachamir Hotza, Sergio Yesid Gómez González.
- Effect of Ni/Mn cationic disorder on the structural and magnetic properties of La₂NiMnO₆, La₂NiFeO₆, and La₂NiMnCoO₆ double perovskites synthesized by sol-gel for hydrogen photoreforming (in preparation). Diego Rueda, Dachamir Hotza, Sergio Yesid Gómez González.
- Evidence of multifunctional properties of ferromanganite La_{0.5}Ca_{0.5}Mn_{0.5}Fe_{0.5}O₃, *Journal of Molecular Structure* 1314, 138821.
- Low-cost tangerine peel waste mediated production of Titanium Dioxide Nanocrystals: Synthesis and characterization, *Environmental Nanotechnology, Monitoring and Management* 13, 100285.
- Spatial and temporal variation of methane emissions in the coastal Balearic Sea, Western Mediterranean, *Science of The Total Environment* 865, 161249.

Work Experience

Assistant professor [Fundación Agraria de Colombia](#) · Full-Time

Bogotá D.C. Feb. 2024 - Sep. 2024

- Professor in various courses of the Environmental Engineering undergraduate program.

Auxiliary professor [National University of Colombia](#) · Full-Time

Bogotá D.C. Aug. 2023 - Sep. 2024

- Aux. professor in various courses of physics department.

Project Consultant [AUDAMA](#)

Cali, Valle del Cauca Nov. 2021 - May. 2022

- Desing and Definition of technical criteria and terms of reference for the selection of hardware, software, and photovoltaic power systems.

- internship - Laboratory technician - researcher.

Scholarships and Awards

- **Meritorious Master's Thesis** Awarded a Meritorious Thesis for the work titled Crystallographic, Electrical, and Magnetic Characteristics of the Double Perovskite Material LaCaMnFeO₆ for academic excellence.
- **CAPES Scholarship** awarded for academic excellence, with the purpose of pursuing a Ph.D. at a Brazilian institution.
- **Teaching Assistant Grant** for academic excellence, designated for undergraduate physics courses, National University of Colombia, 2023.
- **Academic Excellence Non-Tuition Scholarship**, Master's in Physics, National University of Colombia, 2023.
- **JAE INTRO Scholarship**. Internship at CSIC-IMEDEA, 2020.
- **AUIP Scholarship**. Master degree in Cadiz university.

Participation in Events

- **Communities and evolution of double perovskite in hydrogen production and storage: A Bibliometric Analysis**, presented at CACENQ – Congresso Acadêmico da Pós-Graduação em Engenharia Química, April 11, 2025. Diego Rueda, Dachamir Hotza, Sergio Yesid Gómez González. Graduate Program in Chemical Engineering, Federal University of Santa Catarina (UFSC), Brazil.
- **Structural, compositional, morphologic, and magnetic features of the RCA₂SbFe₂O₉ triple perovskite**, presented at the 7th International Meeting for Researchers in Materials and Plasma Technology (IMRMPT), 2023. (Oral presentation - Conference)
- **Spatial and temporal variation of methane emissions in the coastal area of Majorca and Cabrera**, presented at a conference, 2021. (Oral presentation - Conference)
- **Utilization of mandarin peel residues for the production and characterization of titanium dioxide (TiO₂) nanostructures with potential self-cleaning applications**, presented at the National Congress of Physics Engineering | 1st International Conference on Applied Physics, Engineering Innovation (APEI), 2018. (Oral presentation - Conference)
- **Characterization of TiO₂ from mandarin peel residues for industrial production**, presented at To.Sca.Lat 1.0 (Total Scattering Analysis for NanoScience in Latin America), 2018. (Symposium- Oral presentation)
- **Some applications in Environmental and Agroindustrial Engineering**, presented at the VI National Congress of Physics Engineering | 1st International Conference on Applied Physics, Engineering Innovation (APEI), 2018. (Oral presentation - Conference)
- **Green synthesis of TiO₂ nanoparticles using mandarin (Citrus reticulata) peels**, presented as a poster at the 1st Green Chemistry Symposium, 2016. (Oral presentation - Conference)
- **Materials Modeling and Machine Learning Workshop: Molecules and Solids**, Universidad San Sebastián, Facultad de Ingeniería, sede Patagonia, Puerto Montt, Chile, August 4–6, 2025. (Certificate of Participation; online and in-person formats)
- **Machine Learning para Solución de Problemas Ambientales**, Universidad Nacional de Colombia, virtual modality, Bogotá D.C., July 5–August 16, 2024, 40 hours. (Course approved; certificate issued on September 2, 2024)

Research Projects

- **Production of Biocompatible Nanomaterials Applied to Magnetic Hyperthermia Therapy**, 2020–2024. (Participants: Diego Rueda Cadavid; Doris Cadavid (Responsible); Laura Huertas; Omar Olarte)
- **Synthesis and Characterization of Nanostructures from Agro-industrial Products and Residues**, 2018–2019. (Participants: Diego Rueda Cadavid; Valentina Arias; Ana Cecilia Agudelo (Responsible); Doris Cadavid; Wilson

Daniel Caicedo Chacón; Oscar Danilo Guerra Ceballos; Alejandra Carreño Sánchez; Angélica María Carvajal Soto; Cristhofer David Vidal Rosa; Carlos Agudelo Morales; Brayan Alexis Sánchez Zapata)

- **Synthesis of Selected Nanostructures for Environmental and Agro-industrial Applications**, 2017. (Participants: Diego Rueda Cadavid; Valentina Arias; Ana Cecilia Agudelo (Responsible); Doris Cadavid; Wilson Daniel Caicedo Chacón; Carlos Agudelo Morales)